

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Data Warehouse vs Operational Databases



Learning Outcome:

Designs the data structure based on the database structures and its characteristics.

Understand the concept of data warehouse, its components and characteristics.

Can Do Statements:

- I can explain why data warehouse is different from operational databases.
- I can give and explain examples from both data warehouse and operational databases.

Ask your partner:

- What is an operational database?
- Why data warehouse is different from operational databases?





Let's Explore!

The differences between operational and warehouse databases are...



Instructions: Read the text below.

- Walter:** Laura, how did you do in the test?
- Laura:** I did really well. I got all of the answers right.
- Walter:** Good for you! I didn't have all of the questions right, so I haven't gotten a good grade.
- Laura:** What questions did you get wrong?
- Walter:** Well, one of the questions is why data warehouse is different from operational databases?
- Laura:** That's an easy question. Look, operational databases haven't been developed online, but they have been developed in transaction processing while data warehouse has focused on online analytical processing. In other words, operational databases are ideal to look for particular information since it maintains current data which might be important for current processes. On the other hand, data warehouse displays data in a general way, I mean, it maintains data in a historical way.
- Walter:** I got it! I didn't have those notes on my notebook; so I think I haven't paid attention during classes.
- Laura:** What's the other question?
- Walter:** It says; can you give and explain examples of both data warehouse and operational databases?
- Laura:** This is a little bit more complex, but it's not that difficult. Remember that an operational database is usually process-oriented or process-driven, I mean, it concentrates on particular business processes such as registration or billing.
- Walter:** What about data warehouse?
- Laura:** Data in a warehouse is non-volatile. It means that new data can be incorporated any time is needed without changing any old information. That's why I told you that data warehouse keeps data in a historical way. Whenever you need any information it will be available to only be read.



**Let's Practice!**

Laura explains some examples from data warehouse and operational databases

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: didn't, hasn't, realize, It wasn't, kept.

1. This operational database _____ incorporated data. It wasn't designed with that purpose.
2. Operational database hasn't _____ the old information stored as the data warehouse did it in the past.
3. I _____ know that this operational database hasn't had information available to only be read.
4. I didn't _____ that data in warehouse hasn't processed specific information.
5. Data warehouse hasn't focused on particular business processes yet. _____ thought to be focused on business processes such as registration or billing.

**Let's Practice!**

Laura explains why data warehouse is different from operational databases

Instructions: Match the sentences from Column A with Column B. Read the sentences aloud. Check your pronunciation with your instructor.

COLUMN A

1. Operational databases haven't
2. Operational databases hadn't enough online
3. I didn't know that the data warehouse hasn't been designed
4. The data warehouse hasn't updated current data. It wasn't
5. The data warehouse hasn't displayed data in a specific way. It didn't

COLUMN B

- a. include that option since it was created.
- b. created to develop that kind of processes as the data warehouse did.
- c. tools in the past, but they haven't been developed with virtual tools yet.
- d. been developed online; they weren't designed with that technology as the data warehouse did.
- e. to look for particular information. The data warehouse is in charge of that.



Extension Activity

Now it's your turn to explain examples about data warehouse and operational databases

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find examples about data warehouse and operational databases.
- In a poster, write the most important examples combining words and pictures.
- Give an oral presentation to explain the examples written in the poster.

Self-Assessment

Yes

No

- I find information about data warehouse and operational databases.
- I write in a poster the most important examples.
- I give an oral presentation to explain the most important examples of data warehouse and operational databases.

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Target Words

- **I mean** (expression): indicates that you want to say something in a different way
e.g. On the other hand, data warehouse displays data in a general way, I mean, it maintains data in a historical way.
- **transaction** (noun): a type of interaction between two parties
e.g. Look, operational databases haven't been developed online, but they have been developed in transaction processing while data warehouse has focused on online analytical processing.
- **really** (interjection): exclamation to exalt something
e.g. I did really well. I got all of the answers right.

- **good for you** (expression): indicates approval for something another person does
e.g. Good for you! I didn't have all of the questions right, so I haven't gotten a good grade.
- **particular** (adjective): that is unique
e.g. In other words, operational databases are ideal to look for particular information, since it maintains current data which might be important for current processes.
- **ideal** (adjective): that is the best
e.g. In other words, operational databases are ideal to look for particular information, since it maintains current data which might be important for current processes.



Learning Points

Present perfect vs simple past: negative statements

You can use the **present perfect** and **simple past** at the same time to contrast information between what you did in the past (**simple past**) and what you started in the past and continue doing in the present (**present perfect**).

To create a **negative statement** using **simple past**, you need to use the auxiliary a verb DID and NOT with in its base form.

Structure:

Pronoun + Aux + Not + Verb + Complement

Example: *I **did not know** the differences between operational databases and data warehouse.*

To create a **negative statement** using **present perfect**, you need to use the auxiliaries HAVE (I-you-we-they) and HAS (he-she-it) and NOT with a verb in **past participle** form.

Structure:

Pronoun + Aux + Not + Verb (past participle) + Complement

Example: *Operational databases **haven't been developed** online.*

You can include a **simple past** statement with a **present perfect** statement in one sentence to contrast actions that are completed and actions that have not been completed.

Example:

*I **didn't have** all of the questions right, so I **haven't gotten** a good grade.*

Example

- *I **didn't have** those notes on my notebook; so I think I **haven't paid** attention during classes.*



Learning Points

Operational database vs data warehouse

There are some different characteristics for **operational databases** and **data warehouse**.

Something really important for companies is to handle databases correctly. Data must be available at any time, that's why there are two types of databases; operational and warehouse.

Data warehouse contains all of the data stored from an organization; it basically aims for analyzing information and answer some simple questions as who or what. On the other hand, **operational databases** are refreshed regularly and contains current information.

The main characteristic of these databases is that both have the same goal, which is to save and organize data either in a historical way or an updatable way.